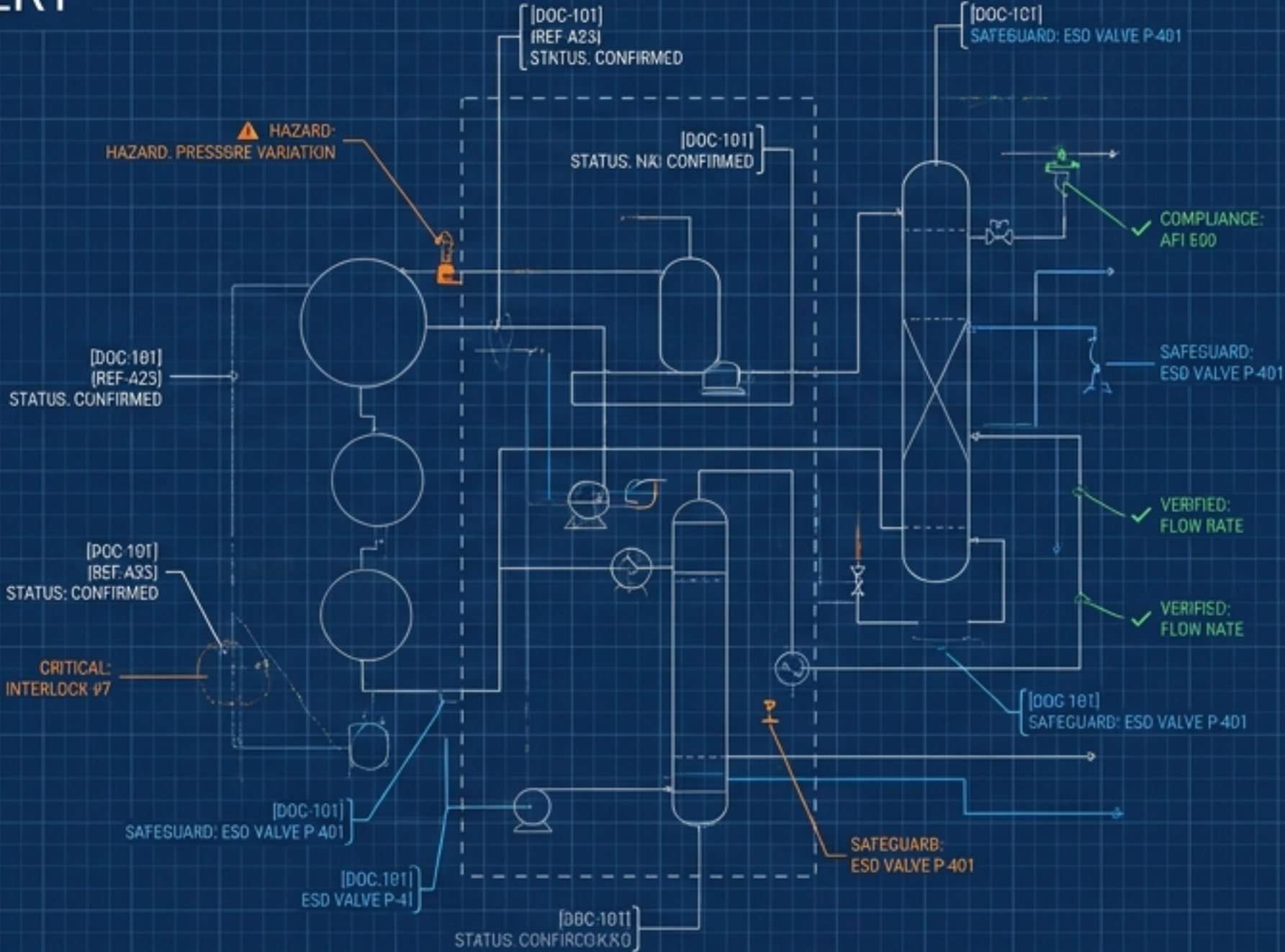


Synthesizing Plant Documentation into Defensible Engineering Intelligence.

SYSTEM INITIATION // PHENOL PROCESS EXPERT



Plant:	PTT Phenol Train II (PPCL), Map Ta Phut, Thailand
Licensors:	UOP
Engineering:	POSCO Engineering
Scope Area:	CDN Section (Concentration / Decomposition / Neutralization)
Date:	2026-06-25

A Wiki-Grounded Expert Assistant, Not a Hallucinating Search Tool.

Standard Chatbot / Search Tool

- ✗ Guesses when data is missing.
- ✗ Performs flat text searches across unconnected files.
- ✗ Improvises safety judgments.

Process Expert Assistant

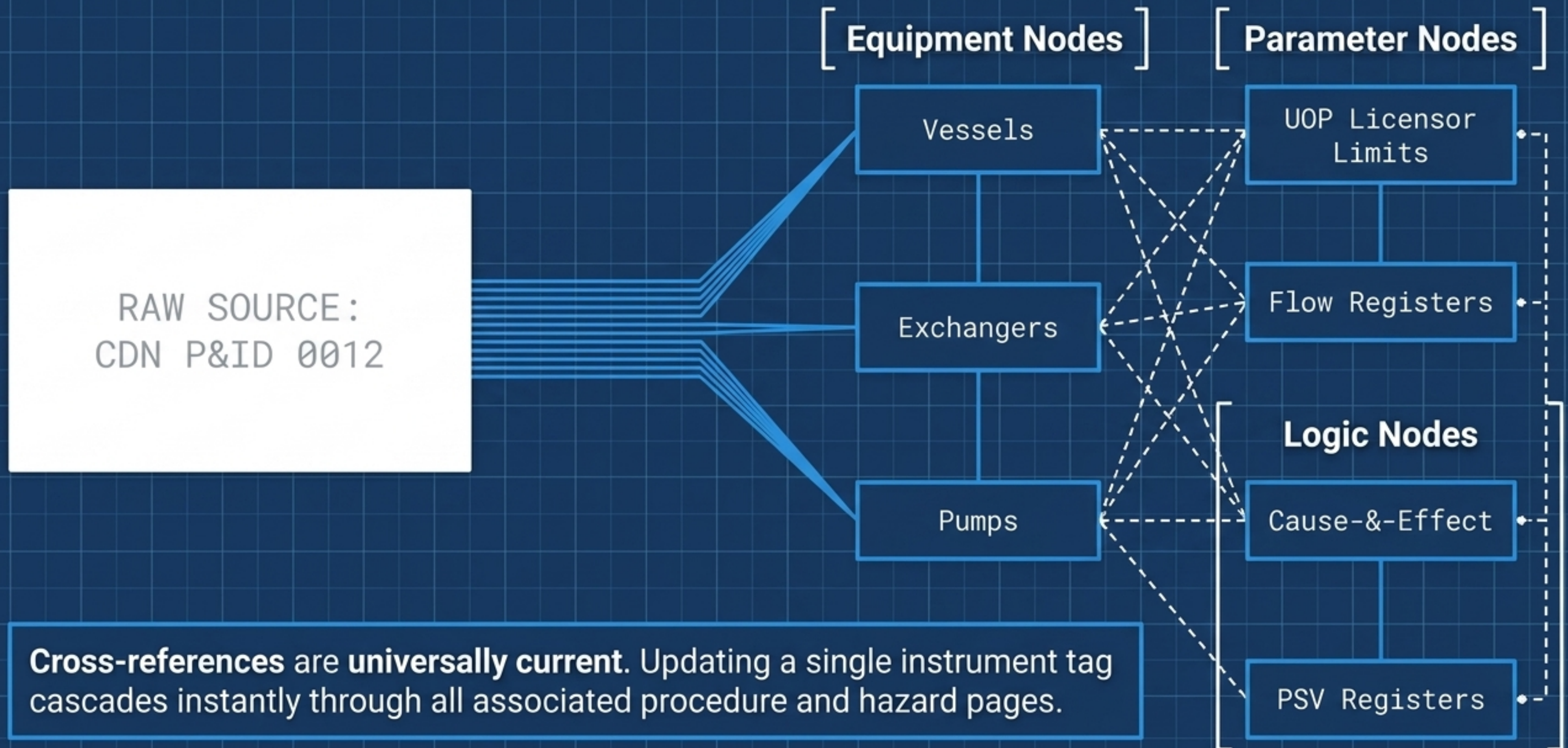
- ✓ Flags gaps and contradicts explicitly when data is missing.
- ✓ Ingests PFDs, P&IDs, data sheets, and manuals into a synthesized, cross-referenced wiki.

Synthesized
Cross-Referenced Wiki

- ✓ **Traces** every single answer directly to a cited source document or wiki page.

[REDACTED] ----- [DOC-101.A PFD REV.4]
[REDACTED] ----- [DOC-202.B P&ID REV.2]
[REDACTED] ----- [DOC-303.C MANUAL REV.1]

One Raw Document Fractures into Dozens of Interconnected Knowledge Nodes.



The Current Knowledge State of the CDN Section

97

Source Documents Ingested

CDN P&IDs 0002-0023, PFDs 0000-0006 fully synthesized.

151+

Synthesized Wiki Pages

54

Equipment Pages

vessels, exchangers, pumps, columns, vacuum & acid systems.

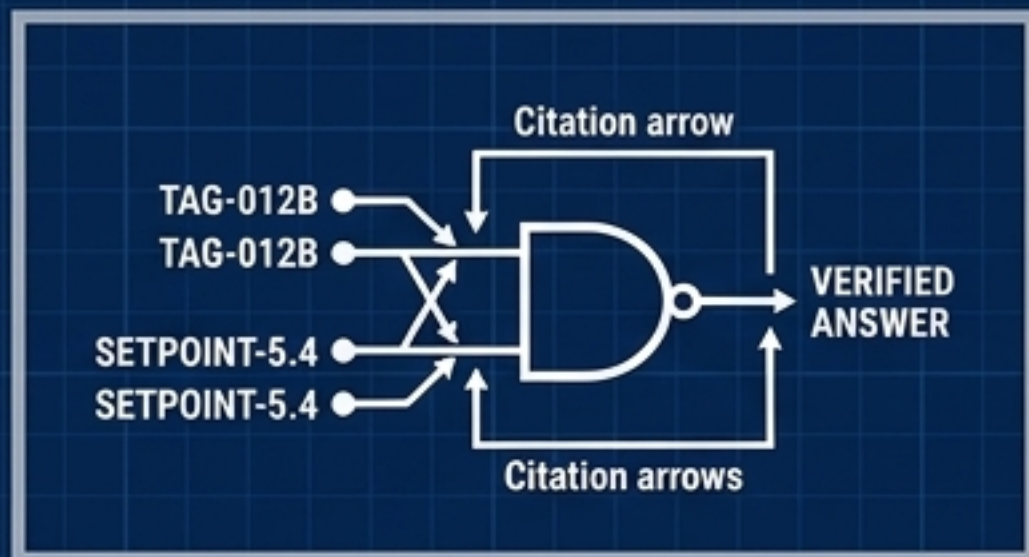
15

GHS Chemical Hazard Pages

CHP, phenol, benzene, cumene, H2SO4.

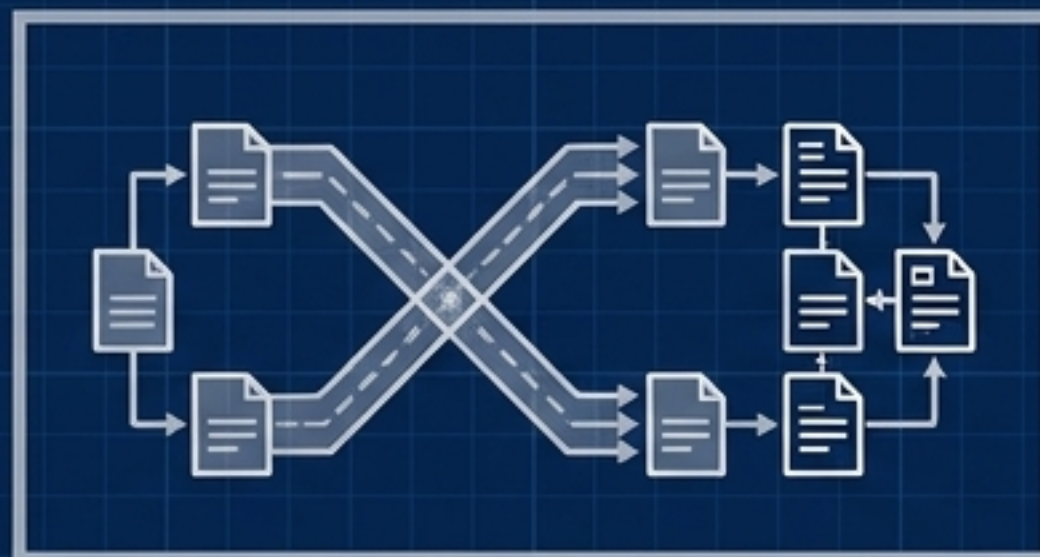
ALKY & OXI equipment captured | Full UOP licensor limits established | Full SIS cause-&-effect and PSV registers mapped

Six Core Execution Capabilities Driven by the Synthesized Wiki.



1. Answer Technical Questions

Grounded, cited, using real setpoints and monospaced tags.



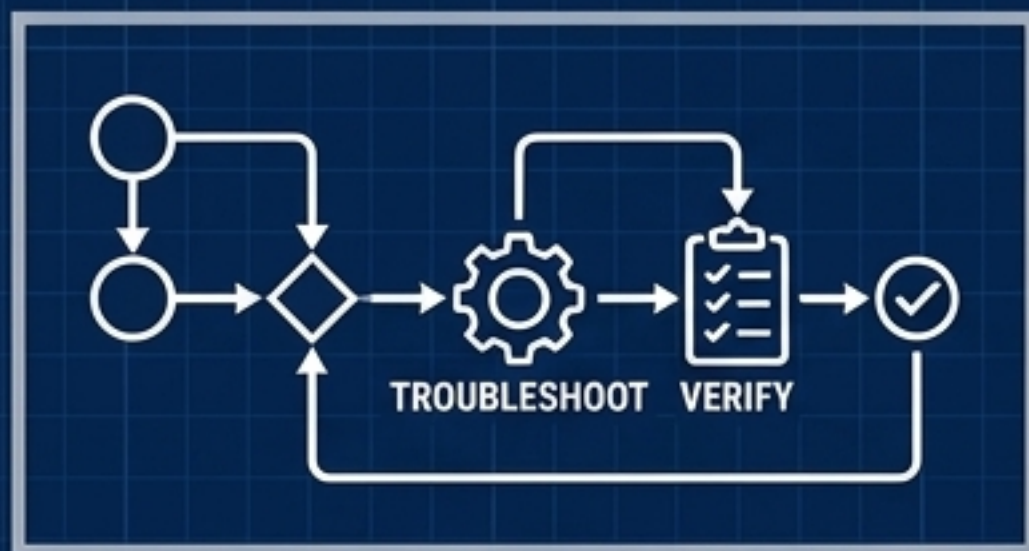
2. Ingest New Documents

Compounding knowledge with automatic cross-referencing.



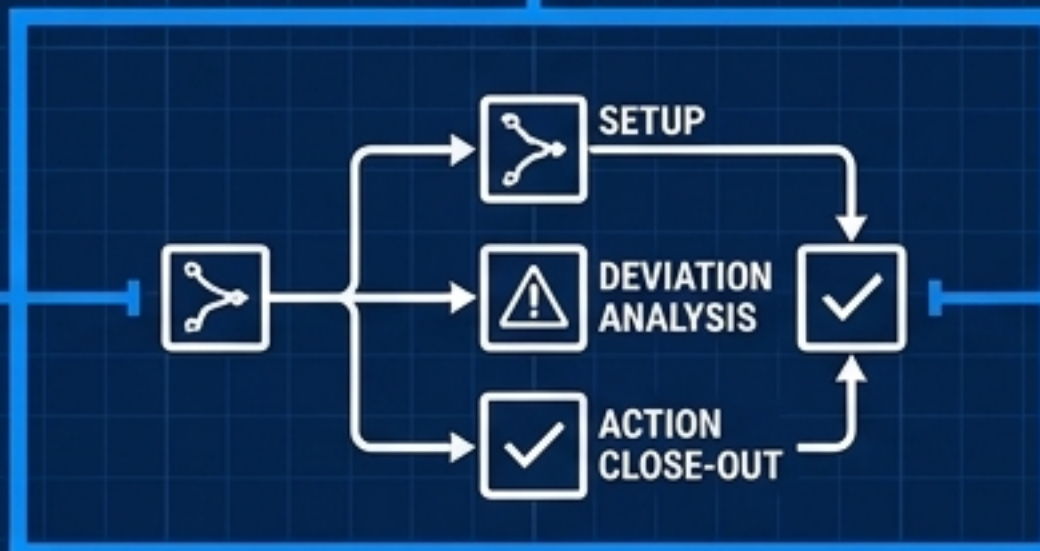
3. Maintain Knowledge Integrity

Active flagging of contradictions, orphan pages, and stale data.



4. Support Engineering Tasks

Instant troubleshooting and procedure verification.



5. Run Full HAZOP Lifecycle

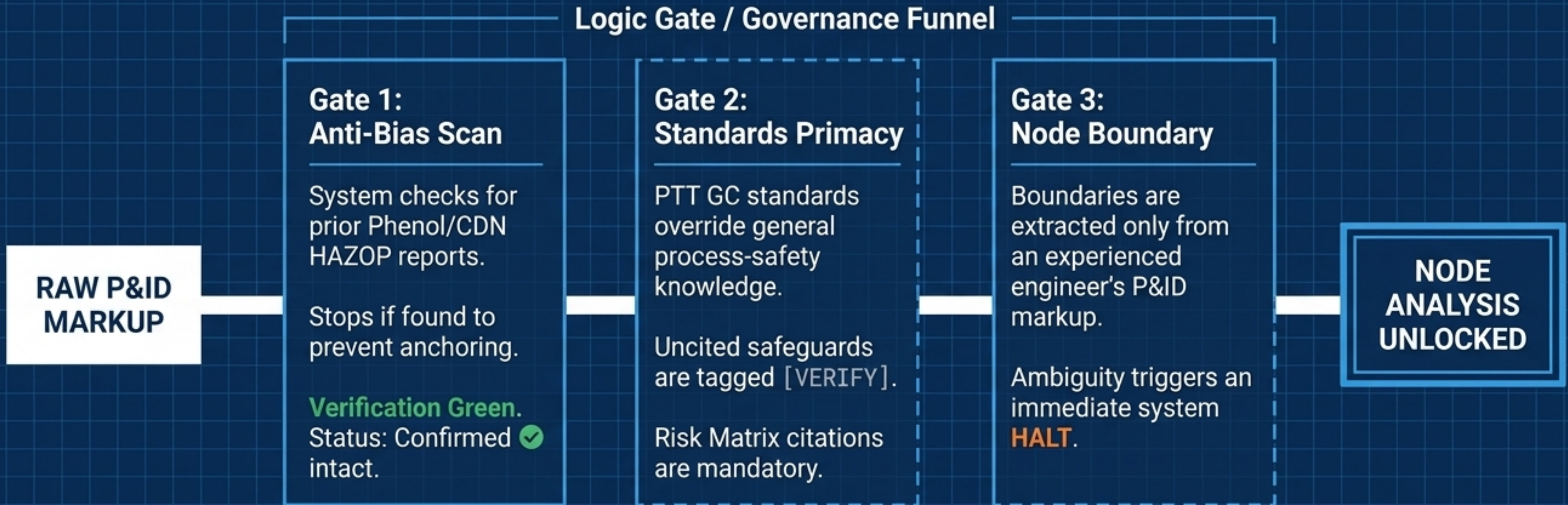
Setup, node deviation analysis, and action close-out.



6. Produce Deliverables

Automated generation of reports, equipment lists, and action registers.

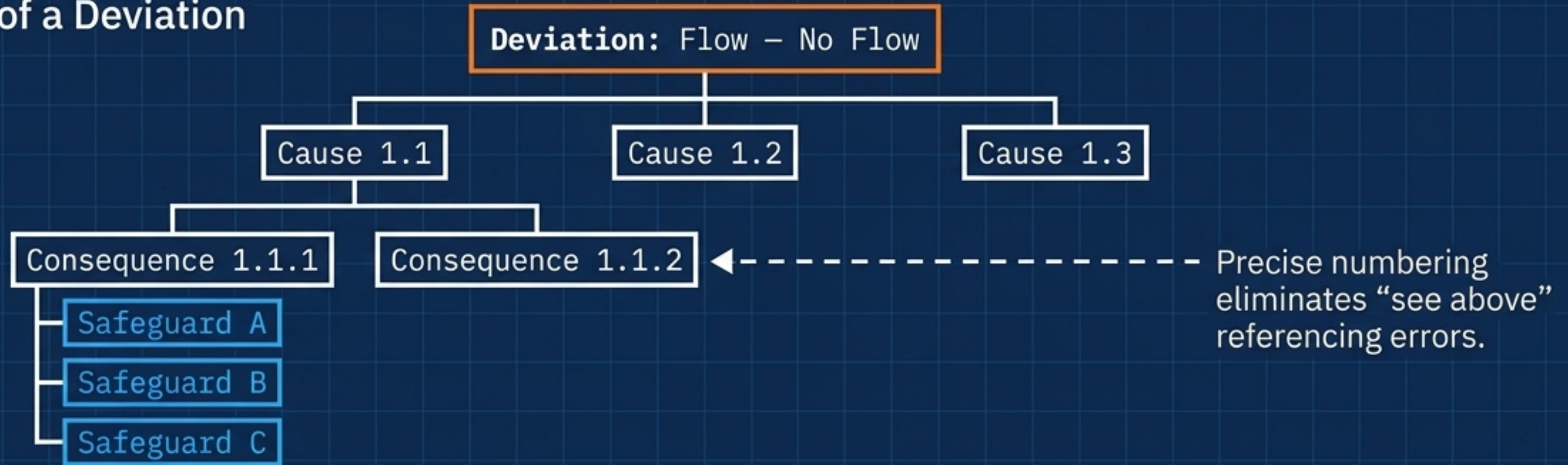
A HAZOP Skill Governed by Three Non-Negotiable Rules



NOTE: The skill is not a prompt—it is the single source of truth for all HAZOP work, enforcing strict procedural discipline.

Generating an Audit-Ready, Hierarchical HAZOP Worksheet.

Anatomy of a Deviation



The Safeguard Lifecycle Timeline



NOTE: Hierarchical structure ensures full traceability and automated report generation, reducing manual effort and errors.

Forcing Granular Specificity and Cross-Node Logic Validation.

Safeguard Specificity

Example Entry:

TXSHH-0701 1oo2 SIL 2 trips UXV-0701

Requires:
Tag +
Setpoint +
Action

Requires:
IL/ESD flag
(real SIS trip
vs. basic alarm)

Requires:
Inline IPL credit level

Governance Cross-Checks

Cross-Node Interlock/ESD Summary

	UXV-0701				
TXSHH-0701			X		

Every credited SIS trip is mathematically checked against the plant's cause-&-effect matrix. ✓

Economic-Severity Conflict Guard

Doc A
(P&ID)

Doc B
(PHA/LOPA)



System detects when two governing docs disagree on Economic severity, stopping analysis to force a "surface-and-ask" before scoring.

What Stands Between 'Equipped' and 'Signed Off' (Honest Status).

High-quality drafts accelerate workshops, but do not replace human team sign-off.
(Drafts CDN-N02, CDN-N03 currently await review).

OPEN ITEM	BLOCKER IMPACT
 [STOP] D-2304 rupture-disc burst pressure conflict	Unresolved sole overpressure protection on the most hazardous CDN vessel.
 [STOP] D-2312 fluid identity (DIAMINE/HMDA vs DIP A)	Unresolved hazard basis for the neutralization node.
 Economic severity category unresolved	Every Economic risk ranking remains provisional until PPCL class is fixed.

Traceability is Defensibility.

A compounding, source-cited process knowledge base that turns months of document hunting into instant, defensible answers—
—driving a standards-compliant HAZOP study from unified data, with the guardrails to keep every result audit-ready.